



Highly enhanced low temperature discharge capacity of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ with lithium boron oxide glass modification



ShuangYuan Tan ^{a, b}, Lei Wang ^a, Liang Bian ^a, JinBao Xu ^{a, *}, Wei Ren ^a, PengFei Hu ^c, AiMin Chang ^a

^a Key Laboratory of Functional Materials and Devices for Special Environments of Chinese Academy of Sciences, Xinjiang Key Laboratory of Electronic Information Materials and Devices, Xinjiang Technical Institute of Physics & Chemistry, Chinese Academy of Sciences, Urumqi 830011, China

^b University of Chinese Academy of Sciences, Beijing 100049, China

^c Laboratory for Microstructures, Shanghai University, Shanghai 200444, China

HIGHLIGHTS

- $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ is modified with lithium boron oxide glass by a solution method.
- At -40°C the discharge capacity of LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ is increased from 37.2 mAh g^{-1} to 101.9 mAh g^{-1} .
- At -40°C , the discharge capacity retention of coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ is 57.8%.

ARTICLE INFO

Article history:

Received 23 August 2014

Received in revised form

26 November 2014

Accepted 30 November 2014

Available online 3 December 2014

Keywords:

Low temperature performance

Glass material

Surface modification

Cathode material

ABSTRACT

Although lithium ion battery is known to be an excellent renewable energy provider in electronic markets further application of it has been limited by its notoriously poor performance at low temperature, especially below -20°C . In this paper, the electrochemical performance of the $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ cathode materials coated by lithium boron oxide (LBO) glass was investigated at a temperature range from 20 to -40°C . The results show that the LBO coating not only helps to improve the discharge capacity of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at room temperature but also increase the discharge capacity retention of the $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ from 22.5% to 57.8% at -40°C . Electrochemical impedance spectra results reveal that the LBO coating plays an important role in reducing the charge-transfer resistance on the electrolyte–electrode interfaces and improving lithium ion diffusion coefficients. The mechanism associated with the change of the structure and electrical properties are discussed in detail.

© 2014 Elsevier B.V. All rights reserved.

1. Introduction

Lithium ion batteries have experienced intensive studies for transportation applications in hybrid electric vehicles (HEV), plug-in hybrid electric vehicles (PHEV), and electric vehicles (EV), being seriously considered for stationary storage and utilization of intermittent renewable energies like solar and wind energies [1,2]. While it is adequate for HEVs and EVs applications under standard lab's environment (room temperature), the discharge capacity of Li-ion batteries is poor at relatively low environmental temperature that HEVs and EVs are supposed to withstand during their normal use.

It has been reported that the malfunction of Li-ion batteries at low temperature is mainly caused by these negative factors including low lithium ion diffusion rate of electrode and electrolyte, high charge-transfer resistance on the electrolyte–electrode interfaces, and high polarization of the electrode [3–7]. In order to solve this issue, there have been many attempts to modify cathode materials like controlling the particle size [8,9], doping with supervalent ion [10] and surface modification [11–15]. Shi et al. [11] used LiF coating to improve cycling performance of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at low temperature, and the discharge capacity was 85.6 mAh g^{-1} for 0.1 C rate at -20°C . Surface modification is an effective and economic way to minimize the direct contact of the active material with electrolyte. Coating can also be adopted to improve ionic or electronic conductivity and optimize the formation of solid–electrolyte interface layer on the cathode surface.

* Corresponding author.

E-mail address: xujb@ms.xjb.ac.cn (J. Xu).

Lithium boron oxide $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ (LBO) glass is a kind of solid lithium ionic conductor (or fast ion conductor) with good lithium ionic conductivity [16–18]. Molten LBO compositions exhibit good wetting properties and relatively low viscosity, which facilitates to form homogeneous coating on the surface [19]. Chan et al. [20,21] and Sahan et al. [22] reported that spinel lithium manganese oxide cathode material coated with LBO exhibits relatively good cycling performance. Choi et al. [23] found that LBO glass-modified LiCoO_2 demonstrated good cycling performance. Chae et al. [24] synthesized a LBO-glass-coated $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ cathode, which exhibits remarkably enhanced electrochemical reversibly and stability at elevated temperature (60 °C). These reports indicated that surface modification with LBO glass on cathode materials can improve their electrochemical performances effectively.

$\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ has been considered to be one of the most attractive cathode materials due to its high capacity, safety and low cost [25]. In our previous work [26], we have found the high rate discharge performance of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ is greatly improved at room temperature by coating with LBO glass layer. In this study the low temperature performance of LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ will be investigated. And the morphology, structure, and electrochemical performances of bare and coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ materials are discussed in detail.

2. Experiment

$\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powder was synthesized via oxalate co-precipitation method [27]. All chemical reagents used in the experiment were of analytical grade (purchased from Aladdin) without further purification. $\text{LiOH}\cdot\text{H}_2\text{O}$, $\text{MnSO}_4\cdot\text{H}_2\text{O}$, $\text{NiSO}_4\cdot 6\text{H}_2\text{O}$, $\text{CoSO}_4\cdot 7\text{H}_2\text{O}$ and $\text{H}_2\text{C}_2\text{O}_4$ were used as raw materials. The mixed solution of metal sulfates (cationic ratio of Ni:Co:Mn = 1:1:1) with total concentration of 2 mol L^{-1} was added drop by drop into the reaction vessel containing oxalic acid solution (used as precipitator), and the co-precipitation reaction was carried out at 50 °C with the stirring speed of 500–750 r/min. During precipitation, the pH value of the reaction bath was adjusted to 8.5 by adding NH_4OH . Later the precursor was centrifuged and dried precipitated and then mixed with stoichiometric amount of $\text{LiOH}\cdot\text{H}_2\text{O}$. The mixture was sintered at 850 °C for 12 h to form $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$.

LBO glass-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ was prepared using a solution method. Stoichiometric $\text{LiOH}\cdot\text{H}_2\text{O}$ and H_3BO_3 (1:2 in molar) was dissolved in deionized water under strong stirring. Bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders were added into the solution stirred continually, and gently heated to 80 °C when the solvent completely evaporated. Then, these powders were calcined at 500 °C for 10 h. After the mixture cooled to room temperature, LBO glass coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders were obtained. The weight ratio of the precursor of LBO glass to $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders for this experiment was 3 wt. %.

The crystal structure was analyzed by X-ray diffraction (XRD, Bruker D8 Advance) with $\text{Cu K}\alpha$ radiation ($\lambda = 1.5418 \text{ \AA}$) under the incidence angle range of 10–80° at a scanning rate of 2° per minute. The particle morphology of bare and coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders was determined by scanning electron microscopy (SEM, Supra55VP, Zeiss). Transmission electron microscope (TEM, JEM-2010, JEOL) was utilized to evaluate the film thickness coated on the surface of cathode powders. Surface properties of sample particles were investigated by X-ray photoelectron spectroscopy (XPS, ESCALAB 250Xi, TMO) with $\text{Al K}\alpha$ radiation. The binding energy was calibrated with reference to C 1s level of carbon (284.8 eV).

Electrochemical characterizations were carried out using the CR2025-type coin cell. The cathode was fabricated from a mixture of the prepared powders, acetylene black, and polyvinylidene fluoride with a weight ratio 8:1:1, and then pasted on aluminum

foil. The pasted cathode foil was cut into circular discs of 10 mm in diameter and dried at 120 °C for 12 h under vacuum. The active materials loading was about 8.2 mg cm^{-2} with a thickness of 80 μm . The cells were assembled in an argon-filled glove-box using lithium metal foil as the counter electrode. The electrolyte was 1 M LiPF_6 in a mixture of ethyl carbonate, propylene carbonate, ethyl methyl carbonate, and butyl acetate. They were charged and discharged between 2.5 and 4.5 V at 0.2 C rate (1 C = 200 mA g^{-1}) with a charge–discharge apparatus (BTS-51, Neware, China). Electrochemical impedance spectroscopy (EIS) was recorded on the electrochemical working station (CHI650, Chenhua, China).

3. Results and discussion

Fig. 1 shows XRD patterns of bare and LBO coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders. Diagrams show well-defined Bragg lines that correspond to the hexagonal $\alpha\text{-NaFeO}_2$ structure ($R\bar{3}m$ space group). Strong and narrow diffraction peaks as well as clear peak splits of (006)/(102) and (018)/(110) reveal the highly ordered layered structure of $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ powders [28]. Compared with bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders, the spectrum of LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ is almost the same and no diffraction peaks of single H_3BO_3 and $\text{LiOH}\cdot\text{H}_2\text{O}$ are observed, which indicates that LBO modification does not destroy the lattice.

In order to confirm further formation of the coating structure in the materials, SEM and TEM were used to study surface morphologies and microstructures of these samples. Firstly it can be seen in Fig. 2 that the bare and coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ samples exhibit similar surface morphologies. The particle size of LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ remains unchanged and there is no agglomeration after modification. Secondly, an obvious thin coating layer with a thickness of approximately 8 nm is observed on the surface of coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ from the TEM images as shown in Fig. 3. This phenomenon suggests that the coating structure has been formed on the surface of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders.

Further evidence for the composition of coating layer is provided by X-ray photoelectron spectroscopy. The results are shown in Figs. 4 and 5. The XPS wide-spectra show that both bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ have Ni 2p, Mn 2p and Co 2p peaks. As shown in Fig. 5 (b–d), the binding energies of Ni $2p_{3/2}$, Mn $2p_{3/2}$, and Co $2p_{3/2}$ for bare/LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ are 855.1/854.8, 642.4/642.0 and 780.0/780.1 eV respectively. These values

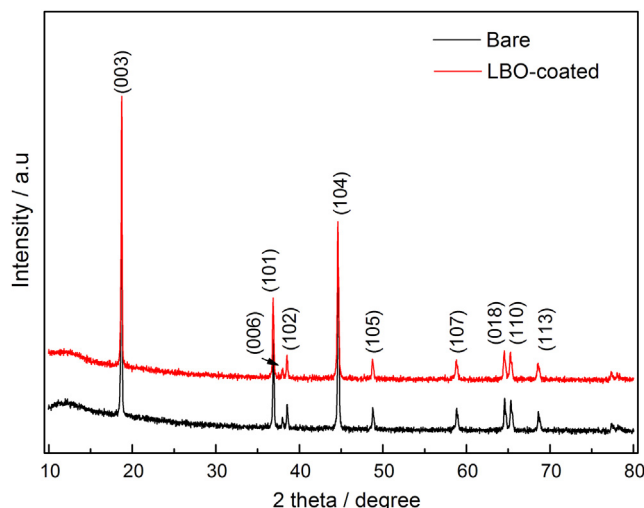


Fig. 1. XRD patterns of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders.

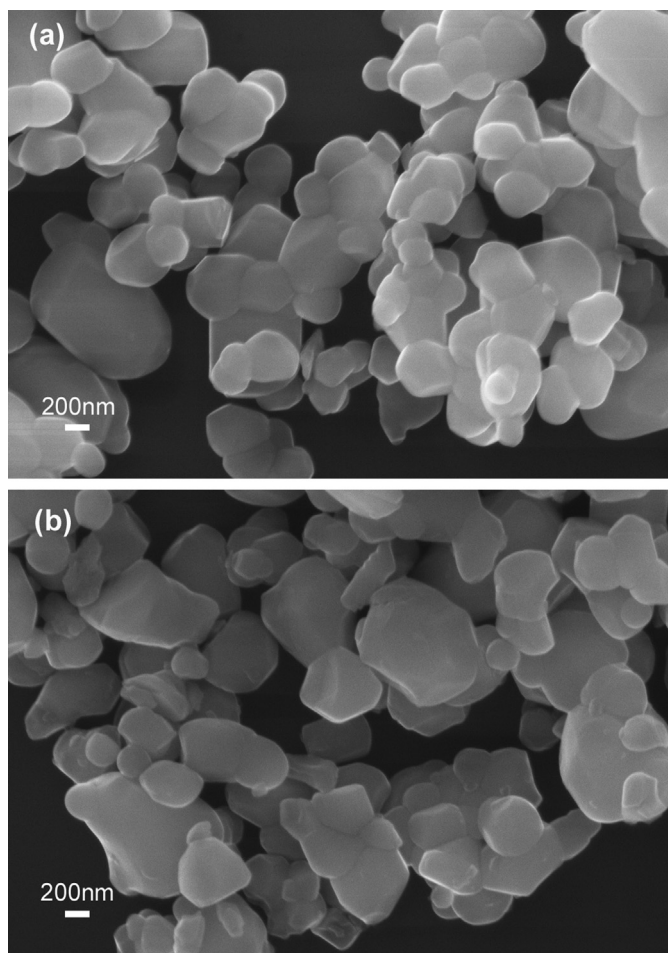


Fig. 2. SEM images of (a) bare and (b) LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$.

suggest that the valences of Ni, Mn, and Co in our sample are divalent, tetravalent, and trivalent, quite consistent with the reported values [29,30]. After coating, peak area ratios of Ni 2p, Mn 2p and Co 2p in Figs. 4 and 5 (b–d) become smaller. The peak area ratios of Ni, Mn and Co between coated and bare samples are 0.338, 0.581 and 0.239 respectively, which indicates the existence of less Ni, Mn and Co on the surface and the environments in the surface

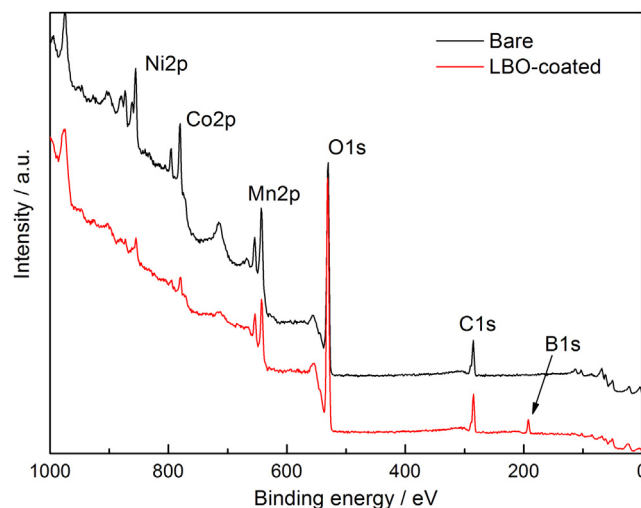


Fig. 4. XPS wide-spectra of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$.

structure for Ni, Mn and Co ion has changed. In Fig. 5e, the shape of O 1s spectra are nearly same, however the amounts of oxygen in two states (529 and 531 eV) are different. The peak around 529 eV can be assigned to the lattice oxygen and another peak around 531 eV is due to the absorbed oxygen. [31] The LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ has more absorbed oxygen and less lattice oxygen on its surface. According to the spectra, there is more oxygen on the surface of coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (the peak area ratio between coated and bare samples is 1.33). The state and amount of oxygen on the surface have been changed after modification. Moreover, boron component is detected in the LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (Figs. 4 and 5a). The observed binding energy of B 1s was 192.3 eV, which is in agreement with the result reported by Choi [23]. All these results suggest that the LBO coating layer has been formed on $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ surface.

The initial charge–discharge curves of bare and coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at 0.2 C rate are shown in Fig. 6. Both samples show a gradually decreasing discharge voltage from 4.5 to 2.5 V, which is the main characteristic of the $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ cathode material. At room temperature, the initial charge/discharge capacities for LBO-coated and bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ are 206.6/175.8 mAh g^{-1} and 208.1/165.4 mAh g^{-1} , respectively. The coulombic efficiency of coated sample is 85.1%, which is higher than

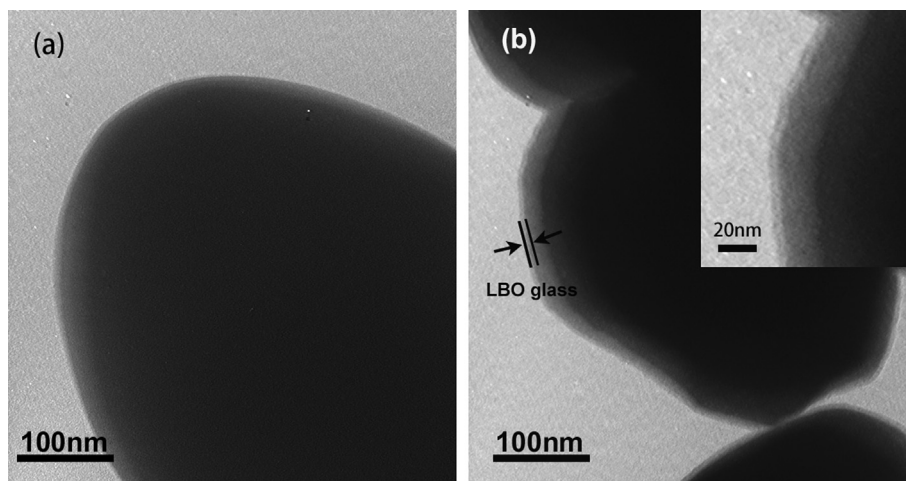


Fig. 3. TEM images of (a) bare and (b) LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$.

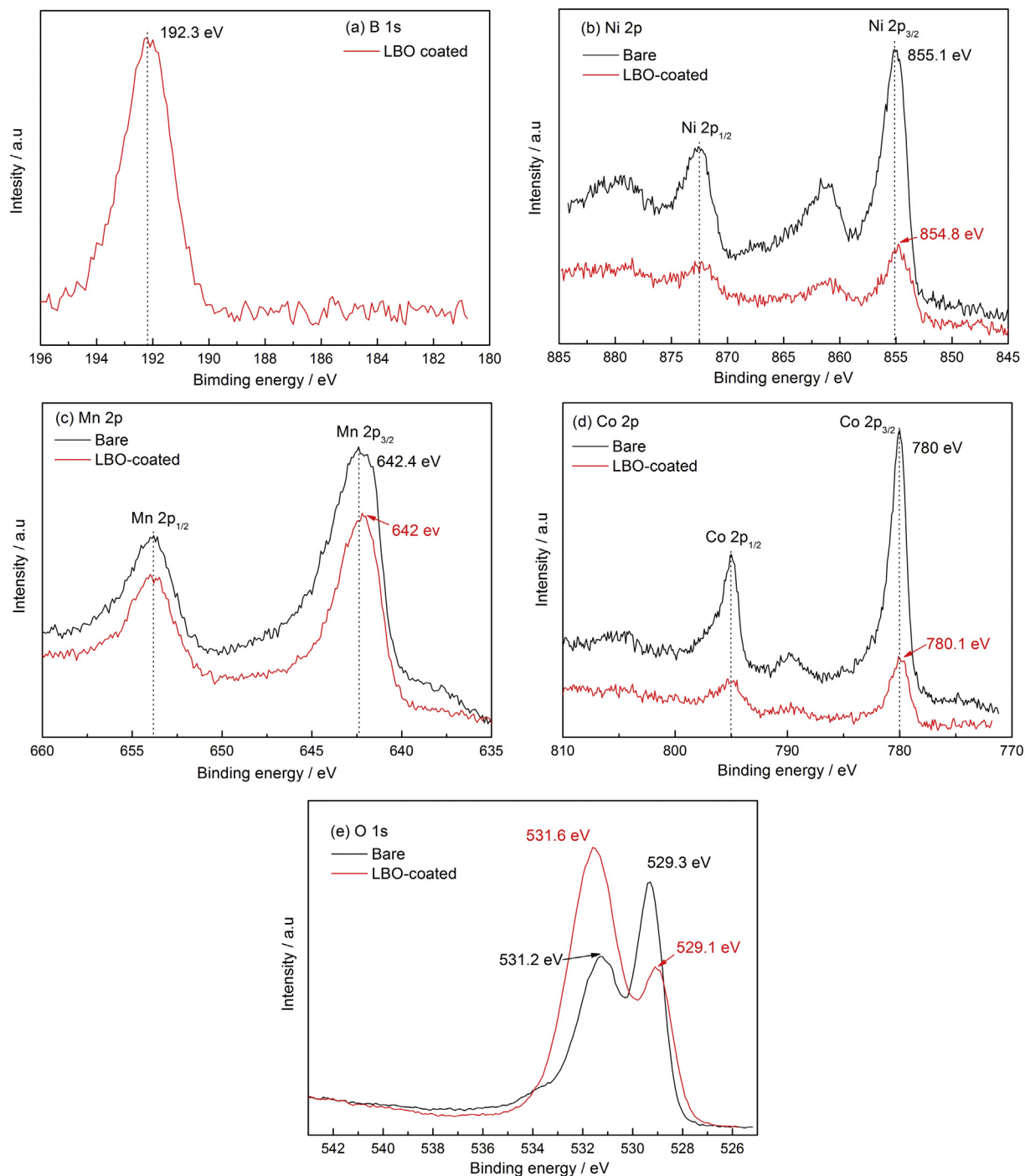


Fig. 5. XPS spectra of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$.

that of the bare sample (79.5%). At the same special capacity, the charge voltage of the coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ is nearly same as that of the bare one, while the discharge voltage is higher. This phenomenon indicates that smaller electrode polarization exists in the coated one. Since the LBO glass has good lithium ionic conductivity, it may enhance the delivery of lithium ions on the surface that in turn contributes to increased capacity of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ cathode material. On the other hand LBO coating layer protects the surface of cathode particles from the liquid electrolyte and unwanted side reactions, thus the charge–discharge polarizations being minimized with LBO-coating.

In order to characterize low temperature properties of these two kinds of samples, the discharge capacity was tested at every 10 °C and the operating temperature range was from room temperature to −40 °C. Discharge curves for bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ cathode materials under a current rate of 0.2 C at different temperature are given in Fig. 7a and b. As the temperature decreases, the discharge capacity of both samples is reduced. The discharge capacity of bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ has a significant decline below −20 °C, in comparison with the smaller capacity decline of coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$. As shown in Fig. 7c, discharge capacity retentions of bare/coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$

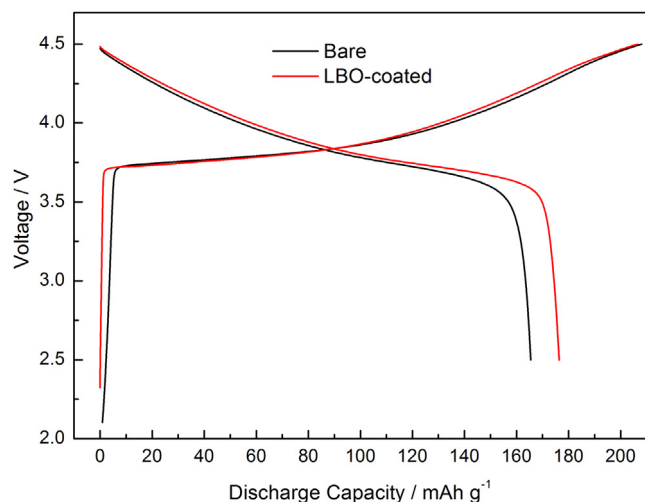


Fig. 6. The initial charge–discharge curves of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ powders at 0.2 C rate.

are nearly the same at $-20 \sim 10^\circ\text{C}$ (they are 90.7/91.9%, 89.1/87.3%, 82.0/82.2% and 72.6/74.8% respectively), but large differences are found below -20°C . At -40°C the discharge capacity of bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ faded from 165.5 mAh g^{-1} to 37.2 mAh g^{-1} with the retention of 22.5% to its initial capacity. However, under the same condition, LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ clearly shows an improved low temperature performance. The discharge capacity of coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ faded from 176.4 mAh g^{-1} to 101.9 mAh g^{-1} at -40°C , with the retention of 57.8% to its initial capacity. The amount of the retention is almost 2.5 times more than that of the bare one.

Fig. 8 shows the cyclic performances of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at 20°C and -20°C . The cells were charged and discharged at 0.2 C in the voltage range of 2.5–4.5 V at same temperature. At room temperature the highest discharge capacity of bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ reaches 169.8 mAh g^{-1} and after 50 cycles the capacity retention is 80.4%. However coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ delivers higher discharge capacity of 185.1 mAh g^{-1} and higher capacity retention of 93.4%, revealing that the cyclic performance is improved by surface coating. As the test temperature falls to -20°C , bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ can deliver a discharge capacity of 70.4 and 103.4 mAh g^{-1} after activation for several cycles [11], respectively. Furthermore coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ exhibits satisfied cyclic performance with capacity retention of 70.6% after 50 cycles, while bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ shows cyclic performance with retention of 28.8% after 50 cycles.

To determine the underlying physical reason for these findings, impedance spectra were measured at different temperatures. Fully discharged coin cells (CR2025, 2.5 V) were measured in the experiment. The Nyquist plots of bare and coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at various temperature are given in Fig. 9. All these curves are composed of a small interrupt, a small semicircle in high frequency, one semicircle in high to medium frequency and an inclined line in low frequency. The small interrupt corresponds to the solution resistance (R_e). The small semicircle in high frequency signifies the diffusion resistance (R_f) of lithium ions through the solid electrolyte interface (SEI) layer. The semicircle in high-to-medium frequency is attributed to the charge transfer resistance (R_{ct}), and the inclined line at low frequency is attributed to Warburg impedance (Z_w) that is associated with lithium diffusion in electrodes [11,32,33]. According to Fig 9, the impedance of the coated sample is smaller than

that of the bare sample. The impedance spectra were fitted with the fitting equivalent circuit as shown in Fig. 10. The fitting results are shown in Table 1.

As shown in Table 1, R_e is the minimum resistance and change a little at different temperature, which means the solution resistance has least effect on low temperature performance. When the temperature decreases, R_f and R_{ct} increase obviously. The decreasing of temperature shows a much more pronounced influence on R_{ct} . R_{ct} corresponding to kinetics of the electrode reaction. Previous researches on the low-temperature performance of Li-ion battery have suggested that R_{ct} and the lithium ion diffusion rate might be the main performance limiting aspects at low temperature [6,34]. The interfacial charge transfer at the interface of electrode and electrolyte is the rate determine step in the Li transfer process. The activation energy (E_a) can be calculated by the temperature dependence charge transfer resistance (R_{ct}) [35]:

$$\ln \frac{T}{R_{ct}} = \ln A - \frac{E_a}{R} \frac{1}{T} \quad (1)$$

where E_a is the activation energy, T is the absolute temperature, R is the gas constant, and A is a temperature-independent constant. The increase in the ionic conductivity may ultimately be connected to a decrease in the activation energies. [36] According to the fitting results in Table 1, the activation energy of bare and coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ can be calculated as $69.14 \text{ kJ mol}^{-1}$ and $65.12 \text{ kJ mol}^{-1}$. The higher E_a means that bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ electrode kinetics is more sluggish. Similar to the reduced trend of discharge capacity, R_{ct} of bare and coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ has significant change between -30°C and -40°C . The above observation further verifies that R_{ct} is one of the key factors that limit the low temperature performance of the cathode material. LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ has much smaller R_{ct} than bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at low temperature. On the other hand, the increase of R_f with the temperature dropping reveals that it is more difficult for lithium ions to pass through SEI layer at low temperature.

The lithium ion diffusion coefficients (D_{Li}) could be calculated by the following equation [14,37]:

$$D_{Li} = \frac{R^2 T^2}{2A^2 n^4 F^4 C^2 \sigma^2} \quad (2)$$

R is the gas constant; T is the absolute temperature; A is the surface area of the cathode; n is the number of electrons per molecule attending the electronic transfer reaction; F is Faraday constant; C is the concentration of lithium ion; and σ is the Warburg factor which has relationship with Z' :

$$Z' = R_e + R_f + R_{ct} + \sigma \omega^{-1/2} \quad (3)$$

Thus, lithium ion diffusion coefficients are inversely proportional to σ^2 . Fig. 11 shows the relationship between Z' and square root of frequency $\omega^{-1/2}$ in low frequency region. These graphs are composed of straight lines and the value of σ could be obtained from the slope. As the temperature drops, the slopes of these lines increase, and show a significant change in the range of -30°C to -40°C . This temperature range is consistent with the critical temperature range in lithium-ion batteries with poor performance. It is found that after LBO coating the Warburg factor has a marked decrease. From these data, we can get $\sigma_{\text{bare}} \approx 1.34\sigma_{\text{coated}}$, $D_{Li, \text{coated}} \approx 1.79D_{Li, \text{bare}}$ at room temperature, revealing that the LBO coating layer improves the Li-ion conductivity of $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$. Furthermore, the improvement is more remarkable at low temperature: at -40°C $\sigma_{\text{bare}} \approx 2.95\sigma_{\text{coated}}$, $D_{Li, \text{coated}} \approx 8.70 D_{Li, \text{bare}}$.

EIS results demonstrate that the amorphous LBO coating layer with high lithium ion ionic conductor on the surface of the $\text{LiNi}_{1/3}$

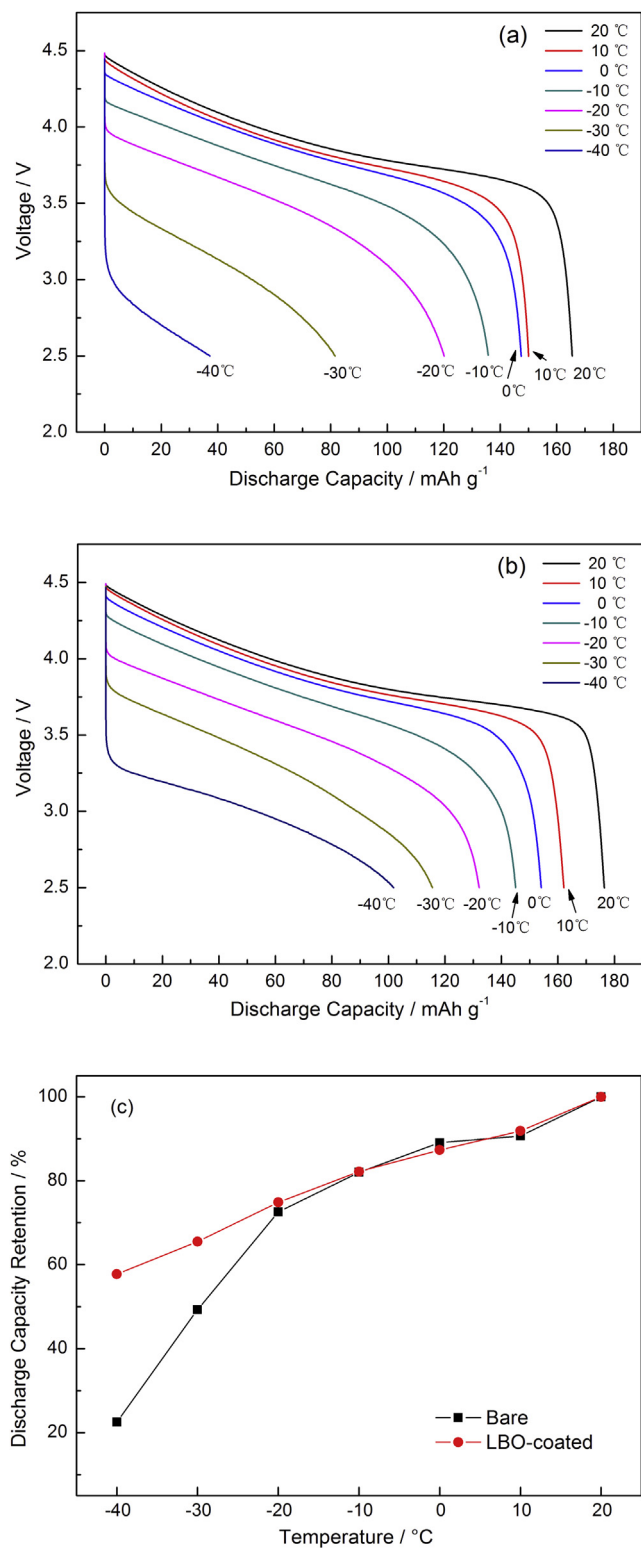


Fig. 7. The discharge curves for (a) bare and (b) LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ under a current rate of 0.2 C at different temperature ($-40 \sim 20^\circ\text{C}$), (c) the discharge capacity retention of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at different temperature ($-40 \sim 20^\circ\text{C}$).

$\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ cathode material can minimize the direct contact of the active material with electrolyte and encourage lithium ion transferring through interfaces between $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ cathode particles. At low temperature, the LBO coating efficiently

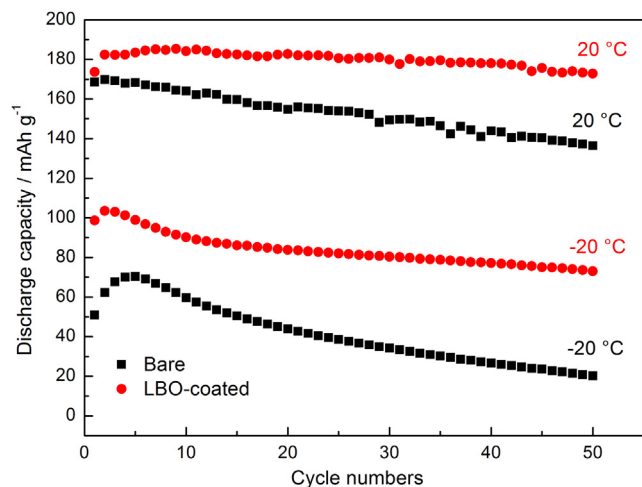


Fig. 8. Cyclic performance of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ at 0.2 C between 2.5 and 4.5 V at 20°C and -20°C .

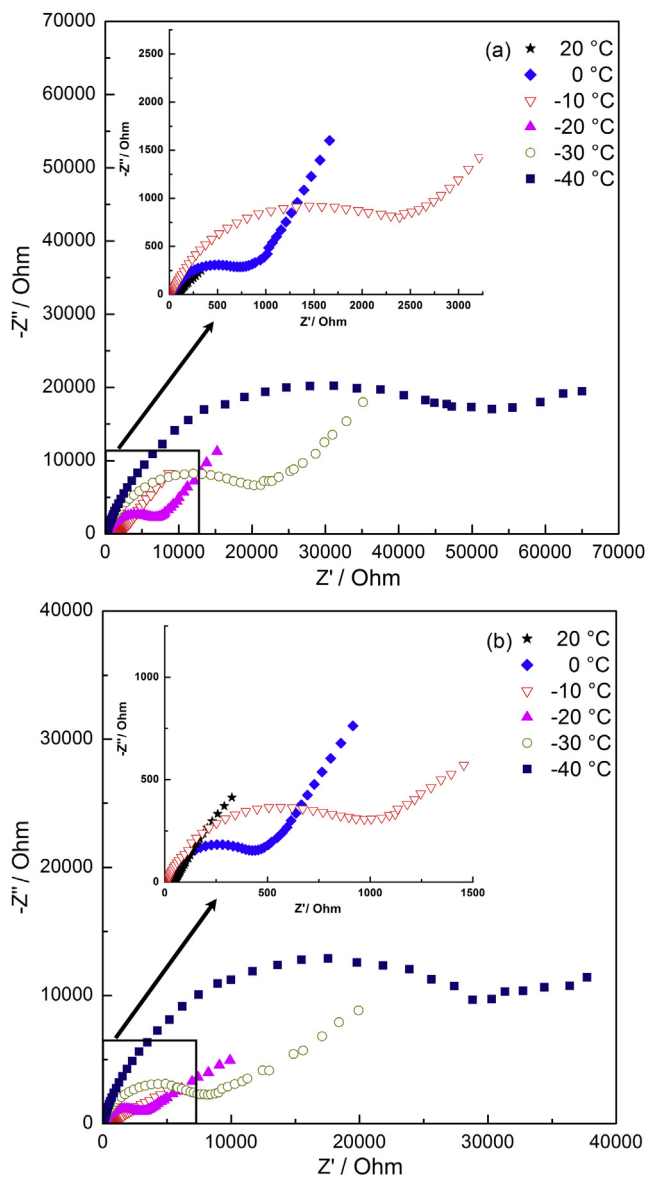


Fig. 9. The Nyquist plots of (a) bare and (b) coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ tested at different temperature.

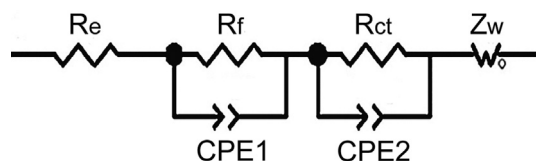


Fig. 10. Equivalent circuit model used for EIS fitting of the electrode.

Table 1

Impedance parameters obtained from equivalent circuit fitting of experimental data for bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ measured at different temperature.

Temperature	Bare			LBO-coated		
	R_e (Ω)	R_f (Ω)	R_{ct} (Ω)	R_e (Ω)	R_f (Ω)	R_{ct} (Ω)
20 °C	6.9	8.4	46.9	6.8	3.3	25.5
0 °C	6.9	30.4	530	7.4	42.8	264.6
-10 °C	8.4	139	1456	11.2	114.1	647.1
-20 °C	11.9	543.7	4731	11.9	523.5	1910
-30 °C	15.7	1924	13,590	13.5	1482	5257
-40 °C	25.5	9008	36,560	21.0	7028	20,300

reduces charge transfer resistance of the Li-ion battery and correspondingly improves ion diffusion coefficients.

4. Conclusion

In conclusion, the discharge capacity of bare and LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ in low temperature environment was

investigated. The coulombic efficiency for initial cycle of the LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ is 85.1% which is higher than that of the bare $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (79.5%) at room temperature and the coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ has smaller electrode polarization. LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ has improved discharge capacity at low temperature especially below -20 °C. At -40 °C the discharge capacity of LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ increases from 37.2 mAh g^{-1} to 101.9 mAh g^{-1} and the discharge capacity retention is almost 2.5 times more than that of the bare one. LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ shows good cyclic performance at both room temperature and -20 °C (with the capacity retention about 93.4% and 70.6% after 50 cycles respectively). Electrochemical impedance spectroscopy shows that the decreased charge transfer resistance and the increased lithium ion diffusion coefficients are main factors that improve the low temperature discharge capacity obviously. The obtained excellent low-temperature discharge capacity of LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ should enable its wide acceptance and application in low-temperature Li-ion batteries.

Acknowledgments

This work was financially supported by the Young Creative Sci-Tech Talents Cultivation Project of Xinjiang Uyghur Autonomous Region (Grant No. 2013731023), West Light Foundation of The Chinese Academy of Sciences (Grant No. XBBS-2014-04), and the National Natural Science Foundation of China (Grant No. 41302029, 41130746).

References

- [1] J.B. Goodenough, J. Solid State Electrochem. 16 (2012) 2019–2029.
- [2] M. Armand, J.M. Tarascon, Nature 451 (2008) 652–657.
- [3] E.J. Plichta, W.K. Behl, J. Power Sources 88 (2000) 192–196.
- [4] C.S. Wang, A.J. Appleby, F.E. Little, J. Electrochem. Soc. 149 (2002) A754–A760.
- [5] J. Fan, S. Tan, J. Electrochem. Soc. 153 (2006) A1081.
- [6] S.S. Zhang, K. Xu, T.R. Jow, J. Power Sources 115 (2003) 137–140.
- [7] J. Yi, U.S. Kim, C.B. Shin, T. Han, S. Park, J. Power Sources 244 (2013) 143–148.
- [8] S.R. Li, Y. Qiao, Y. Sun, S.Y. Ge, Y.M. Chen, I. Lieberwirth, Y. Yu, C.H. Chen, Electrochim. Acta 81 (2012) 191–196.
- [9] S.H. Choi, J.W. Son, Y.S. Yoon, J. Kim, J. Power Sources 158 (2006) 1419–1424.
- [10] C. Li, N. Hua, C. Wang, X. Kang, W. Tuerdi, Y. Han, J. Alloys Compd. 509 (2011) 1897–1900.
- [11] S.J. Shi, J.P. Tu, Y.Y. Tang, Y.Q. Zhang, X.Y. Liu, X.L. Wang, C.D. Gu, J. Power Sources 225 (2013) 338–346.
- [12] Z. Ma, G. Shao, X. Wang, J. Song, G. Wang, T. Liu, J. Solid State Electrochem. 17 (2013) 2409–2416.
- [13] G. Huang, W. Li, H. Sun, J. Wang, J. Zhang, H. Jiang, F. Zhai, Electrochim. Acta 97 (2013) 92–98.
- [14] X.Z. Liao, Z.F. Ma, Q. Gong, Y.S. He, L. Pei, L.J. Zeng, Electrochem. Commun. 10 (2008) 691–694.
- [15] M. Gaberscek, R. Dominko, J. Jamnik, Electrochem. Commun. 9 (2007) 2778–2783.
- [16] E.E. Horopanitis, G. Perentzis, E. Pavlidou, L. Papadimitriou, Ionics 9 (2003) 88–94.
- [17] Y. Ito, K. Miyauchi, T. Oi, J. Non Cryst. Solids 57 (1983) 389–400.
- [18] M. Eddir, P. Dzwonkowski, C. Julien, M. Balkanski, Solid State Ion. 45 (1991) 77–82.
- [19] G.G. Amatucci, A. Blyr, C. Sigala, P. Alfonse, J.M. Tarascon, Solid State Ion. 104 (1997) 13–25.
- [20] H.W. Chan, J.G. Duh, S.R. Sheen, Electrochim. Acta 51 (2006) 3645–3651.
- [21] H.W. Chan, J.G. Duh, S.R. Sheen, Surf. Coat. Technol. 188–189 (2004) 116–119.
- [22] H. Sahan, Solid State Ion. 178 (2008) 1837–1842.
- [23] S.H. Choi, J.H. Kim, Y.N. Ko, K.M. Yang, Y.C. Kang, J. Power Sources 244 (2013) 129–135.
- [24] J.S. Chae, S.B. Yoon, W.S. Yoon, Y.M. Kang, S.M. Park, J.W. Lee, K.C. Roh, J. Alloys Compd. 601 (2014) 217–222.
- [25] T. Ohzuku, Y. Makimura, Chem. Lett. (2001) 642–643.
- [26] J. Dou, X. Kang, T. Wumaier, H. Yu, N. Hua, Y. Han, G. Xu, J. Solid State Electrochem. 16 (2011) 1481–1486.
- [27] T.H. Cho, Y. Shiosaki, H. Noguchi, J. Power Sources 159 (2006) 1322–1327.
- [28] T. Ohzuku, A. Ueda, M. Nagayama, J. Electrochem. Soc. 140 (1993) 1862–1870.
- [29] Y.K. Sun, S.T. Myung, B.C. Park, J. Prakash, I. Belharouak, K. Amine, Nat. Mater. 8 (2009) 320–324.
- [30] D. Li, Y. Kato, K. Kobayakawa, H. Noguchi, Y. Sato, J. Power Sources 160 (2006) 1342–1348.

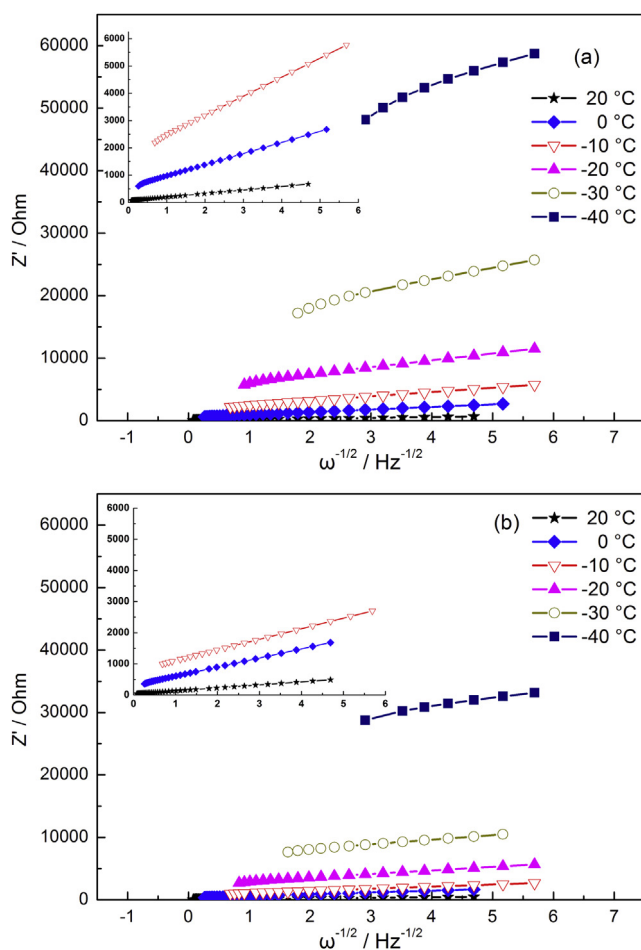


Fig. 11. The relationship between Z' (ohm) and $\omega^{-1/2}$ ($\text{Hz}^{-1/2}$) for (a) bare and (b) LBO-coated $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$.

- [31] J. Gurgul, M.T. Rinke, I. Schellenberg, R. Pöttgen, *Solid State Sci.* 17 (2013) 122–127.
- [32] L. Liu, K. Sun, N. Zhang, T. Yang, J. *Solid State Electrochem.* 13 (2008) 1381–1386.
- [33] T. Liu, S.X. Zhao, K. Wang, C.W. Nan, *Electrochim. Acta* 85 (2012) 605–611.
- [34] C.K. Huang, J.S. Sakamoto, J. Wolfenstine, S. Surampudi, *J. Electrochem. Soc.* 147 (2000) 2893–2896.
- [35] X. Liu, H. Li, E. Yoo, M. Ishida, H. Zhou, *Electrochim. Acta* 83 (2012) 253–258.
- [36] M. Okubo, Y. Tanaka, H.S. Zhou, T. Kudo, I. Honma, *J. Phys. Chem. B* 113 (2009) 2840–2847.
- [37] Z. Liu, X. Kang, C. Li, N. Hua, T. Wumair, Y. Han, *J. Solid State Electrochem.* 16 (2011) 1917–1923.